

Internal Dependencies

References

- [Analyze java package metrics in a graph database](#)
- [Calculate metrics](#)
- [Neo4j Python Driver](#)

Artifacts

List the artifacts this notebook is based on. Different sorting variations help finding artifacts by their features and support larger code bases where the list of all artifacts gets too long.

Only the top 30 entries are shown. The whole table can be found in the following CSV report:

`List_all_Java_artifacts`

Table 1a - Top 30 artifacts with the highest package count

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-messaging-5.0.3.jar	59	579	7	2
1	axon-common-5.0.3.jar	15	156	10	0
2	axon-eventsourcing-5.0.3.jar	7	100	3	4
3	axon-modelling-5.0.3.jar	7	92	2	3
4	axon-spring-boot-autoconfigure-5.0.3.jar	7	75	1	7
5	axon-test-5.0.3.jar	6	78	1	3
6	axon-conversion-5.0.3.jar	5	35	4	1
7	axon-server-connector-5.0.3.jar	5	72	1	4
8	axon-update-5.0.3.jar	5	23	1	1
9	axon-metrics-micrometer-5.0.3.jar	3	16	0	3
10	axon-tracing-opentelemetry-5.0.3.jar	1	5	0	2

Table 1b - Top 30 artifacts with the highest type count

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-messaging-5.0.3.jar	59	579	7	2
1	axon-common-5.0.3.jar	15	156	10	0
2	axon-eventsourcing-5.0.3.jar	7	100	3	4
3	axon-modelling-5.0.3.jar	7	92	2	3
4	axon-test-5.0.3.jar	6	78	1	3
5	axon-spring-boot-autoconfigure-5.0.3.jar	7	75	1	7
6	axon-server-connector-5.0.3.jar	5	72	1	4
7	axon-conversion-5.0.3.jar	5	35	4	1
8	axon-update-5.0.3.jar	5	23	1	1
9	axon-metrics-micrometer-5.0.3.jar	3	16	0	3
10	axon-tracing-opentelemetry-5.0.3.jar	1	5	0	2

Table 1c - Top 30 artifacts with the highest number of incoming dependencies

The following table lists the top 30 artifacts that are used the most by other artifacts (highest count of incoming dependencies, highest in-degree).

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-common-5.0.3.jar	15	156	10	0
1	axon-messaging-5.0.3.jar	59	579	7	2
2	axon-conversion-5.0.3.jar	5	35	4	1
3	axon-eventsourcing-5.0.3.jar	7	100	3	4
4	axon-modelling-5.0.3.jar	7	92	2	3
5	axon-server-connector-5.0.3.jar	5	72	1	4
6	axon-spring-boot-autoconfigure-5.0.3.jar	7	75	1	7
7	axon-test-5.0.3.jar	6	78	1	3
8	axon-update-5.0.3.jar	5	23	1	1
9	axon-metrics-micrometer-5.0.3.jar	3	16	0	3
10	axon-tracing-opentelemetry-5.0.3.jar	1	5	0	2

Table 1d - Top 30 artifacts with the highest number of outgoing dependencies

The following table lists the top 30 artifacts that are depending on the highest number of other artifacts (highest count of outgoing dependencies, highest out-degree).

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-spring-boot-autoconfigure-5.0.3.jar	7	75	1	7
1	axon-eventsourcing-5.0.3.jar	7	100	3	4
2	axon-server-connector-5.0.3.jar	5	72	1	4
3	axon-metrics-micrometer-5.0.3.jar	3	16	0	3
4	axon-modelling-5.0.3.jar	7	92	2	3
5	axon-test-5.0.3.jar	6	78	1	3
6	axon-messaging-5.0.3.jar	59	579	7	2
7	axon-tracing-opentelemetry-5.0.3.jar	1	5	0	2
8	axon-conversion-5.0.3.jar	5	35	4	1
9	axon-update-5.0.3.jar	5	23	1	1
10	axon-common-5.0.3.jar	15	156	10	0

Table 1e - Top 30 artifacts with the lowest package count

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-tracing-opentelemetry-5.0.3.jar	1	5	0	2
1	axon-metrics-micrometer-5.0.3.jar	3	16	0	3
2	axon-conversion-5.0.3.jar	5	35	4	1
3	axon-server-connector-5.0.3.jar	5	72	1	4
4	axon-update-5.0.3.jar	5	23	1	1
5	axon-test-5.0.3.jar	6	78	1	3
6	axon-eventsourcing-5.0.3.jar	7	100	3	4
7	axon-modelling-5.0.3.jar	7	92	2	3
8	axon-spring-boot-autoconfigure-5.0.3.jar	7	75	1	7
9	axon-common-5.0.3.jar	15	156	10	0
10	axon-messaging-5.0.3.jar	59	579	7	2

Table 1f - Top 30 artifacts with the lowest type count

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-tracing-opentelemetry-5.0.3.jar	1	5	0	2
1	axon-metrics-micrometer-5.0.3.jar	3	16	0	3
2	axon-update-5.0.3.jar	5	23	1	1
3	axon-conversion-5.0.3.jar	5	35	4	1
4	axon-server-connector-5.0.3.jar	5	72	1	4
5	axon-spring-boot-autoconfigure-5.0.3.jar	7	75	1	7
6	axon-test-5.0.3.jar	6	78	1	3
7	axon-modelling-5.0.3.jar	7	92	2	3
8	axon-eventsourcing-5.0.3.jar	7	100	3	4
9	axon-common-5.0.3.jar	15	156	10	0
10	axon-messaging-5.0.3.jar	59	579	7	2

Table 1g - Top 30 artifacts with the lowest number of incoming dependencies

The following table lists the top 30 artifacts that are used the least by other artifacts (lowest count of incoming dependencies, lowest in-degree).

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-metrics-micrometer-5.0.3.jar	3	16	0	3
1	axon-tracing-opentelemetry-5.0.3.jar	1	5	0	2
2	axon-server-connector-5.0.3.jar	5	72	1	4
3	axon-spring-boot-autoconfigure-5.0.3.jar	7	75	1	7
4	axon-test-5.0.3.jar	6	78	1	3
5	axon-update-5.0.3.jar	5	23	1	1
6	axon-modelling-5.0.3.jar	7	92	2	3
7	axon-eventsourcing-5.0.3.jar	7	100	3	4
8	axon-conversion-5.0.3.jar	5	35	4	1
9	axon-messaging-5.0.3.jar	59	579	7	2
10	axon-common-5.0.3.jar	15	156	10	0

Table 1h - Top 30 artifacts with the lowest number of outgoing dependencies

The following table lists the top 30 artifacts that are depending on the lowest number of other artifacts (lowest count of outgoing dependencies, lowest out-degree).

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-common-5.0.3.jar	15	156	10	0
1	axon-conversion-5.0.3.jar	5	35	4	1
2	axon-update-5.0.3.jar	5	23	1	1
3	axon-messaging-5.0.3.jar	59	579	7	2
4	axon-tracing-opentelemetry-5.0.3.jar	1	5	0	2
5	axon-metrics-micrometer-5.0.3.jar	3	16	0	3
6	axon-modelling-5.0.3.jar	7	92	2	3
7	axon-test-5.0.3.jar	6	78	1	3
8	axon-eventsourcing-5.0.3.jar	7	100	3	4
9	axon-server-connector-5.0.3.jar	5	72	1	4
10	axon-spring-boot-autoconfigure-5.0.3.jar	7	75	1	7

Cyclic Dependencies

Cyclic dependencies occur when one package uses a class of another package and vice versa. These dependencies can lead to problems when one of these packages needs to be changed.

Table 2a - Cyclic Dependencies Overview

Show the top 40 cyclic dependencies sorted by the most promising to resolve first. This is done by calculating the number of forward dependencies (first cycle participant to second cycle participant) in relation to backward dependencies (second cycle participant back to first cycle participant). The higher this rate (approaching 1), the easier it should be to resolve the cycle by focussing on the few backward dependencies.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

[Cyclic_Dependencies](#)

Columns:

- *artifactName* identifies the artifact of the first participant of the cycle
- *packageName* identifies the package of the first participant of the cycle
- *dependentArtifactName* identifies the artifact of the second participant of the cycle
- *dependentPackageName* identifies the package of the second participant of the cycle
- *forwardToBackwardBalance* is between 0 and 1. High for many forward and few backward dependencies.
- *numberForward* contains the number of dependencies from the first participant of the cycle to the second one
- *numberBackward* contains the number of dependencies from the second participant of the cycle back to the first one
- *someForwardDependencies* lists some forward dependencies in the text format "type1 -> type2"
- *backwardDependencies* lists the backward dependencies in the format "type1 <- type2" that are recommended to get resolved

	artifactName	packageName	dependentArtifactName	dependentPackageName	forwardToBackwardBalance	numberForward	numberBack
0	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	0.959184	48	
1	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling	axon-messaging-5.0.3	org.axonframework.messaging.core	0.942857	34	
2	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	0.931034	28	
3	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	0.875000	15	
4	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	0.857143	13	
5	axon-modelling-5.0.3	org.axonframework.modelling.annotation	axon-modelling-5.0.3	org.axonframework.modelling	0.846154	12	
6	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork.transaction	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork	0.666667	5	
7	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.processing.streaming.pooled	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.configuration	0.666667	15	
8	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.processing.subscribing	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.configuration	0.666667	15	
9	axon-common-5.0.3	org.axonframework.common.configuration	axon-common-5.0.3	org.axonframework.common.infra	0.529412	13	
10	axon-messaging-5.0.3	org.axonframework.messaging.core.interception.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	0.500000	15	
11	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.event	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	0.428571	5	
12	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.command	0.333333	4	
13	axon-messaging-5.0.3	org.axonframework.messaging.core.configuration	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.configuration	0.333333	2	
14	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling	axon-messaging-5.0.3	org.axonframework.messaging.core.sequencing	0.333333	4	
15	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.query	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	0.200000	6	
16	axon-modelling-5.0.3	org.axonframework.modelling.entity.annotation	axon-modelling-5.0.3	org.axonframework.modelling.annotation	0.200000	3	
17	axon-modelling-5.0.3	org.axonframework.modelling.entity.child	axon-modelling-5.0.3	org.axonframework.modelling.entity	0.142857	8	
18	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork	axon-messaging-5.0.3	org.axonframework.messaging.core	0.111111	10	
19	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.util	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	0.000000	2	

	artifactName	packageName	dependentArtifactName	dependentPackageName	forwardToBackwardBalance	numberForward	numberBack
20	axon-eventsourcing-5.0.3	org.axonframework.eventsourcing.annotation.reflection	axon-eventsourcing-5.0.3	org.axonframework.eventsourcing.annotation	0.000000	1	

Table 2b - Cyclic Dependencies Break Down

Lists packages with cyclic dependencies with every dependency in a separate row sorted by the most promising dependency first.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

[Cyclic_Dependencies_Breakdown](#)

Columns in addition to Table 2a:

- *dependency* shows the cycle dependency in the text format "type1 -> type2" (forward) or "type2<-type1" (backward)

[illegible]

Table 2c - Cyclic Dependencies Break Down - Backward Dependencies Only

Lists packages with cyclic dependencies with every dependency in a separate row sorted by the most promising dependency first. This table only contains the backward dependencies from the second participant of the cycle back to the first one that are the most promising to resolve.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Cyclic_Dependencies_Breakdown_BackwardOnly

	artifactName	packageName	dependentArtifactName	dependentPackageName	dependency forName
0	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	HandlerAttributes<-SimpleHandlerAttributes
1	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling	axon-messaging-5.0.3	org.axonframework.messaging.core	EventMessage<-SubscribableEventSource
2	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	EventHandler<-HandlerTypeResolver
3	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	CommandHandler<-HandlerTypeResolver
4	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	QueryHandler<-HandlerTypeResolver
5	axon-modelling-5.0.3	org.axonframework.modelling.annotation	axon-modelling-5.0.3	org.axonframework.modelling	TargetEntityIdMemberMismatchException<-PropertyBasedEntityIdResolver
6	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork.transaction	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork	TransactionManager<-TransactionalUnitOfWorkFactory
7	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.processing.streaming.pooled	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.configuration	PooledStreamingEventProcessorsConfigurer<-EventProcessingConfigurer
8	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.processing.streaming.pooled	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.configuration	PooledStreamingEventProcessorConfiguration<-EventProcessorModule
9	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.processing.streaming.pooled	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.configuration	PooledStreamingEventProcessorModule<-EventProcessorModule
10	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.processing.subscribing	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.configuration	SubscribingEventProcessorModule<-EventProcessorModule
11	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.processing.subscribing	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.configuration	SubscribingEventProcessorsConfigurer<-EventProcessingConfigurer
12	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.processing.subscribing	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.configuration	SubscribingEventProcessorConfiguration<-EventProcessorModule
13	axon-common-5.0.3	org.axonframework.common.configuration	axon-common-5.0.3	org.axonframework.common.infra	Component<-JacksonComponentDescriptor
14	axon-common-5.0.3	org.axonframework.common.configuration	axon-common-5.0.3	org.axonframework.common.infra	Component\$Identifier<-JacksonComponentDescriptor
15	axon-common-5.0.3	org.axonframework.common.configuration	axon-common-5.0.3	org.axonframework.common.infra	Component<-FileSystemStyleComponentDescriptor
16	axon-common-5.0.3	org.axonframework.common.configuration	axon-common-5.0.3	org.axonframework.common.infra	Component\$Identifier<-FileSystemStyleComponentDescriptor
17	axon-messaging-5.0.3	org.axonframework.messaging.core.interception.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	NoMoreInterceptors<-ChainedMessageHandlerInterceptorMember
18	axon-messaging-5.0.3	org.axonframework.messaging.core.interception.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	MessageHandlerInterceptorMemberChain<-AnnotatedHandlerInspector
19	axon-messaging-5.0.3	org.axonframework.messaging.core.interception.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	NoMoreInterceptors<-AnnotatedHandlerInspector
20	axon-messaging-5.0.3	org.axonframework.messaging.core.interception.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	MessageInterceptingMember<-AnnotatedHandlerInspector
21	axon-messaging-5.0.3	org.axonframework.messaging.core.interception.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	MessageHandlerInterceptorMemberChain<-ChainedMessageHandlerInterceptorMember
22	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.event	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	EventProcessorControlService<-AxonServerConfigurationEnhancer
23	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.event	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	AxonServerEventStorageEngineFactory<-AxonServerConfigurationEnhancer
24	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.command	MetadataConverter<-CommandConverter
25	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.command	AxonServerConfiguration<-AxonServerCommandBusConnector
26	axon-messaging-5.0.3	org.axonframework.messaging.core.configuration	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.configuration	MessagingConfigurer<-EventProcessingConfigurer
27	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling	axon-messaging-5.0.3	org.axonframework.messaging.core.sequencing	EventMessage<-ExtractionSequencingPolicy
28	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling	axon-messaging-5.0.3	org.axonframework.messaging.core.sequencing	EventMessage<-SequentialPerAggregatePolicy
29	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.query	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	AxonServerRemoteQueryHandlingException<-ErrorCode
30	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.query	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	AxonServerNonTransientRemoteQueryHandlingException<-ErrorCode
31	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.query	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	AxonServerQueryDispatchException<-ErrorCode
32	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.query	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	AxonServerQueryBusConnector<-AxonServerConfigurationEnhancer
33	axon-modelling-5.0.3	org.axonframework.modelling.entity.annotation	axon-modelling-5.0.3	org.axonframework.modelling.annotation	AnnotatedEntityMetamodel<-EntityIdResolverDefinition
34	axon-modelling-5.0.3	org.axonframework.modelling.entity.annotation	axon-modelling-5.0.3	org.axonframework.modelling.annotation	AnnotatedEntityMetamodel<-AnnotationBasedEntityIdResolverDefinition
35	axon-modelling-5.0.3	org.axonframework.modelling.entity.child	axon-modelling-5.0.3	org.axonframework.modelling.entity	EntityChildMetamodel<-ConcreteEntityMetamodel
36	axon-modelling-5.0.3	org.axonframework.modelling.entity.child	axon-modelling-5.0.3	org.axonframework.modelling.entity	ChildAmbiguityException<-ConcreteEntityMetamodel
37	axon-modelling-5.0.3	org.axonframework.modelling.entity.child	axon-modelling-5.0.3	org.axonframework.modelling.entity	EntityChildMetamodel<-PolymorphicEntityMetamodel\$Builder
38	axon-modelling-5.0.3	org.axonframework.modelling.entity.child	axon-modelling-5.0.3	org.axonframework.modelling.entity	EntityChildMetamodel<-PolymorphicEntityMetamodel\$Builder
39	axon-modelling-5.0.3	org.axonframework.modelling.entity.child	axon-modelling-5.0.3	org.axonframework.modelling.entity	EntityChildMetamodel<-ConcreteEntityMetamodel\$Builder

Interface Segregation Candidates

Well known from [Design Principles and Design Patterns by Robert C. Martin](#), the *Interface Segregation Principle* suggests that software components should have narrow, focused interfaces rather than large, general-purpose ones. The goal is to minimize the dependencies between components and increase modularity, flexibility, and maintainability.

Smaller, focused and purpose-driven interfaces

- make it easier to modify individual components without affecting the rest of the system.
- make it clearer which client is affected by which change.
- don't force their clients to depend on methods they don't need.
- reduce the scope of changes since a change to one component doesn't affect others.
- lead to a more loosely coupled architecture that is easier to understand and maintain.

Reference: [Analyze java package metrics in a graph database](#)

How to apply the results

If just one method of a type is used, especially in many places, then the result of this method can be used to call e.g. a method or construct an object instead of using the whole object and then just calling that single method.

If there are a couple of methods that are used for a distinct purpose, those could be factored out into a separate interface. The original type can extended/implement the new interface so that there are no breaking changes. Then all the callers, that use only this group of methods, can be changed to the new interface.

Table 4 - Top 40 most used combinations of methods

The following table shows the top 40 most used combinations of methods of larger types that might benefit from applying the *Interface Segregation Principle*. The whole table can be found in the CSV report `Candidates_for_Interface_Segregation`.

	fullDependentTypeName	declaredMethods	calledMethodNames	calledMethods	callerTypes
0	org.axonframework.messaging.core.unitofwork.ProcessingContext	32	[computeResourceIfAbsent]	1	7
1	org.axonframework.messaging.commandhandling.CommandBus	6	[dispatch]	1	7
2	org.axonframework.common.configuration.ComponentDefinition\$ComponentCreator	17	[createComponent]	1	5
3	org.axonframework.messaging.core.MessageStream\$Entry	7	[message]	1	5
4	org.axonframework.messaging.core.unitofwork.ProcessingContext	32	[withResource]	1	4
5	org.axonframework.messaging.eventhandling.EventMessage	19	[identifier, timestamp]	2	4
6	org.axonframework.messaging.eventhandling.EventMessage	19	[timestamp]	1	4
7	org.axonframework.messaging.core.MessageStream\$Empty	44	[cast]	1	3
8	org.axonframework.messaging.core.DelayedMessageStream	42	[create]	1	3
9	org.axonframework.messaging.eventhandling.EventMessage	19	[andMetadata]	1	3
10	org.axonframework.messaging.core.annotation.WrappedMessageHandlingMember	13	[handleSync]	1	3
11	org.axonframework.messaging.core.unitofwork.UnitOfWork	24	[execute]	1	2
12	org.axonframework.test.fixture.AxonTestThenMessage	22	[exceptionSatisfies, exception]	3	2
13	org.axonframework.messaging.commandhandling.CommandMessage	20	[routingKey, priority]	2	2
14	org.axonframework.messaging.commandhandling.CommandMessage	19	[withConvertedPayload]	1	2
15	org.axonframework.messaging.commandhandling.CommandMessage	19	[andMetadata]	1	2
16	org.axonframework.messaging.eventhandling.EventMessage	19	[withConvertedPayload]	1	2
17	org.axonframework.messaging.eventhandling.EventMessage	19	[identifier]	1	2
18	org.axonframework.messaging.eventstreaming.EventCriteria	12	[tags]	1	2
19	org.axonframework.messaging.eventstreaming.OrEventCriteria	12	[or]	1	2
20	org.axonframework.modelling.entity.ConcreteEntityMetamodel	11	[forEntityClass]	1	2
21	org.axonframework.modelling.entity.PolymorphicEntityMetamodel	11	[forSuperType]	1	2
22	org.axonframework.event sourcing.eventstore.EventStore	9	[transaction]	1	2
23	org.axonframework.messaging.eventhandling.processing.streaming.token.WrappedToken	9	[unwrapLowerBound]	1	2
24	org.axonframework.messaging.eventhandling.processing.streaming.token.store.ConfigToken	9	[get]	1	2
25	org.axonframework.messaging.core.MessageStream\$Entry	7	[map]	1	2
26	org.axonframework.conversion.ChainingContentTypeConverter	5	[canConvert, registerConverter]	2	2
27	org.axonframework.messaging.core.EmptyMessageStream	44	[instance]	1	1
28	org.axonframework.event sourcing.eventstore.inmemory.InMemoryEventStorageEngine\$MapBackedMessageStream	43	[isCompleted, hasNextAvailable]	2	1
29	org.axonframework.event sourcing.eventstore.inmemory.InMemoryEventStorageEngine\$MapBackedMessageStream	42	[callback]	1	1
30	org.axonframework.messaging.core.DelayedMessageStream	42	[createSingle]	1	1
31	org.axonframework.messaging.core.MessageStream\$Single	42	[asCompletableFuture]	1	1
32	org.axonframework.messaging.core.MessageStream\$Single	42	[first]	1	1
33	org.axonframework.messaging.core.MessageStream\$Single	42	[cast]	1	1
34	org.axonframework.messaging.core.unitofwork.ProcessingContext	37	[removeResource, putResourceIfAbsent, updateResource, putResource, computeResourceIfAbsent]	6	1
35	org.axonframework.messaging.core.unitofwork.ProcessingContext	36	[updateResource, computeResourceIfAbsent]	2	1
36	org.axonframework.messaging.core.unitofwork.ProcessingContext	35	[computeResourceIfAbsent]	1	1
37	org.axonframework.messaging.core.unitofwork.ProcessingContext	34	[computeResourceIfAbsent]	1	1
38	org.axonframework.messaging.core.unitofwork.ProcessingContext	33	[putResource]	1	1
39	org.axonframework.messaging.core.unitofwork.ProcessingContext	33	[removeResource, computeResourceIfAbsent]	2	1

Package Usage

Table 5 - Types that are used by multiple packages

This table shows the top 40 packages that are used by the highest number of different packages. The whole table can be found in the CSV report `List_types_that_are_used_by_many_different_packages` .

	fullQualifiedDependentTypeName	dependentTypeName	dependentTypeLabels	numberOfUsingPackages
0	org.axonframework.messaging.core.unitofwork.ProcessingContext	ProcessingContext	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mar...	59
1	org.axonframework.common.annotation.Internal	Internal	[Type, File, Java, ByteCode, Annotation, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent...	55
2	org.axonframework.messaging.core.Message	Message	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mar...	47
3	org.axonframework.common.infra.ComponentDescriptor	ComponentDescriptor	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0...	46
4	org.axonframework.messaging.core.MessageStream	MessageStream	[Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWea...	36
5	org.axonframework.messaging.eventhandling.EventMessage	EventMessage	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponen...	35
6	org.axonframework.messaging.core.MessageType	MessageType	[Type, File, Java, ByteCode, Record, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0...	31
7	org.axonframework.messaging.core.QualifiedName	QualifiedName	[Type, File, Java, ByteCode, Record, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4T...	30
8	org.axonframework.common.configuration.Configuration	Configuration	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation9, Mark4TypeLouvainCommunity5...	29
9	org.axonframework.common.FutureUtils	FutureUtils	[Type, File, Java, ByteCode, Class, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation8, Mark4TypeLouvainCommunity4, Mark4L...	28
10	org.axonframework.messaging.core.Context\$ResourceKey	Context\$ResourceKey	[Type, File, Java, ByteCode, GenericDeclaration, Class, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyCon...	25
11	org.axonframework.messaging.core.MessageStream\$Empty	MessageStream\$Empty	[Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation12, Mark4TypeLouvainCommunity8, Mark4TypeLeidenCommunity9, Mark4TypeK...	25
12	org.axonframework.common.infra.DescribableComponent	DescribableComponent	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0...	24
13	org.axonframework.common.Assert	Assert	[Type, File, Java, ByteCode, Class, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation5, Mark4TypeLouvainCommunity0, Mark4TypeLeidenCommunity10, Mark4TypeCoreDecompositi...	22
14	org.axonframework.common.configuration.ComponentRegistry	ComponentRegistry	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation9, Mark4TypeLouvainCommunity5, Mark4TypeLeidenCommunity6, Mark4TypeCoreDecomposition10, Mark4TypeMaximumKCut18, Mark4TypeHDBSCAN63]	22
15	org.axonframework.common.BuilderUtils	BuilderUtils	[Type, File, Java, ByteCode, Class, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation22, Mark4TypeLouvainCommunity12, Mark4TypeLeidenCommunity13, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut0, Mark4TypeHDBSCAN41, Mark4TopAnomalyHub]	21
16	org.axonframework.messaging.commandhandling.CommandMessage	CommandMessage	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity2, Mark4TypeLeidenCommunity2, ...	20
17	org.axonframework.messaging.core.annotation.ParameterResolverFactory	ParameterResolverFactory	[Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation9, Mark4TypeLouvainCommunity11, Mark4TypeLeidenCommunity3, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut89, Mark4TypeHDBSCAN97]	20
18	org.axonframework.messaging.core.MessageStream\$Entry	MessageStream\$Entry	[Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0,	19

	fullQualifiedDependentTypeName	dependentTypeName	dependentTypeLabels	numberOfUsingPackages
			Mark4TypeLabelPropagation12, Mark4TypeLouvainCommunity8, Mark4T...	
19	org.axonframework.common.configuration.ConfigurationEnhancer	ConfigurationEnhancer	[Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation9, Mark4TypeLouvainCommunity5, Mark4TypeLeidenCommunity6, Mark4TypeCoreDecomposition8, Mark4TypeMaximumKCut0, Mark4TypeHDBSCAN133]	18
20	org.axonframework.conversion.Converter	Converter	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation7, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity1, Mark4TypeCoreDecompo...	18
21	org.axonframework.messaging.core.MessageStream\$Single	MessageStream\$Single	[Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation12, Mar...	18
22	org.axonframework.messaging.core.MessageTypeResolver	MessageTypeResolver	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation7, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity1, Mark4TypeCoreDecomposition10, ...	18
23	org.axonframework.messaging.core.conversion.MessageConverter	MessageConverter	[Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation9, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity6, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut38, Mark4TypeLocalClusteringCoefficient0.11333333333333333, Mark4TypeHDBSCAN111]	17
24	org.axonframework.messaging.core.Metadata	Metadata	[Type, File, Java, ByteCode, Class, Mark4TopCentralityPageRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation0, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity1, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut0, Mar...	16
25	org.axonframework.common.ObjectUtils	ObjectUtils	[Type, File, Java, ByteCode, Class, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation7, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity1, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut0, Mark4TypeHDBSCAN-1]	16
26	org.axonframework.messaging.eventhandling.processing.streaming.token.TrackingToken	TrackingToken	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation15, Mark4TypeLouvainCommunity4...	15
27	org.axonframework.common.AxonConfigurationException	AxonConfigurationException	[Type, File, Java, ByteCode, Class, Throwable, Mark4TopCentralityBetweenness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation30, Mark4TypeLouvainCommunity3, Mark4TypeLeidenCommunity3, Mark4TypeCoreDecomposition8, Mark4TypeMaximumKCut1, Mark4TypeHDBSCAN54]	14
28	org.axonframework.messaging.commandhandling.CommandBus	CommandBus	[Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity2, Mark4TypeLeidenCommunity2, Mark4TypeCoreDecomposition10, Mark4TypeMaximumKCut24, Mark4TypeHDBSCAN71]	13
29	org.axonframework.common.ReflectionUtils	ReflectionUtils	[Type, File, Java, ByteCode, Class, Mark4TopCentralityBetweenness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation5, Mark4TypeLouvainCommunity0, Mark4TypeLeidenCommunity3, Mark4TypeCoreDecomposition8, Mark4TypeMaximumKCut27, Mark4TypeHDBSCAN49, Mark4TopAnomalyBottleneck, Mark4TopA...	13
30	org.axonframework.messaging.commandhandling.CommandResultMessage	CommandResultMessage	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity2, Mark4TypeLeidenCommunity2, Mark4TypeCoreDecomposition10, ...	12
31	org.axonframework.common.configuration.ComponentDefinition	ComponentDefinition	[Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation9, Mark4TypeLouvainCommunity10, Mark4TypeLeidenCommunity6, Mark4TypeCoreDecomposition10, Mark4TypeMaximumKCut19, Mark4TypeHDBSCAN-1]	12
32	org.axonframework.common.configuration.ComponentDefinition\$IncompleteComponentDefinition	ComponentDefinition\$IncompleteComponentDefinition	[Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation9, Mark4TypeLouvainCommunity10, Mark4TypeLeidenCommunity6, Mark4TypeCoreDecomposition10, Mark4TypeMaximumKCut20, Mark4TypeHDBSCAN-1]	12
33	org.axonframework.messaging.core.Context	Context	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation12, Mark4TypeLouvainCommunity8, Mark4TypeLeidenC...	12
34	org.axonframework.messaging.queryhandling.QueryMessage	QueryMessage	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation62, Mark4TypeLouvainCommunity8, Mark4TypeLeidenCommunity9, Mark4TopAnomalyBottleneck, Mark4TopA...	12
35	org.axonframework.common.AxonNonTransientException	AxonNonTransientException	[Type, File, Java, ByteCode, Class, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation5, Mark4TypeLouvainCommunity6, Mark4TypeLeidenCommunity7, Mark4TypeCoreDecomposition4, Mark4TypeMaximumKCut24, Mark4TypeLocalClusteringCoef...	11
36	org.axonframework.common.configuration.ComponentBuilder	ComponentBuilder	[Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation9, Mark4TypeLouvainCommunity5, Mark4TypeLeidenCommunity6, ...	11

	fullQualifiedDependentTypeName	dependentTypeName	dependentTypeLabels	numberOfUsingPackages
			Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut45, Mark4TypeHDBSCAN125]	
37	org.axonframework.messaging.eventhandling.conversion.EventConverter	EventConverter	[Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation9, Mark4TypeLouvainCommunity11, Mark4TypeLeidenCommunity2, Mark4TypeKCoreDecomposition9, Mark4TypeMaximumKCut28, Mark4TypeHDBSCAN147]	11
38	org.axonframework.messaging.core.MessageHandlerInterceptor	MessageHandlerInterceptor	[Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation9, Mark4TypeLouvainCommunity5, Mark4TypeLeidenCommunity6, Mark4TypeKC...	11
39	org.axonframework.messaging.core.annotation.ParameterResolver	ParameterResolver	[Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation7, Mark4TypeLouvainCommunity3, Mark4TypeLeidenCommunity3, Mark4TypeKCoreDecomposition7, Mark4TypeMaximumKCut90, Mark4TypeHDBSCAN128]	11

Table 6 - Packages that are used by multiple artifacts

This table shows the top 30 artifacts that only use a few (compared to all existing) packages of another artifact. The whole table can be found in the CSV report `ArtifactPackageUsage`.

	artifactName	dependentArtifactName	dependentPackages	dependentArtifactPackages	packageUsagePercentage	dependentFullQualifiedPackageNames	dependentPackageNames
0	axon-tracing-opentelemetry-5.0.3	axon-messaging-5.0.3	2	59	0.033898	[org.axonframework.messaging.core, org.axonframework.messaging.tracing]	[core, tracing]
1	axon-tracing-opentelemetry-5.0.3	axon-common-5.0.3	1	15	0.066667	[org.axonframework.common]	[common]
2	axon-metrics-micrometer-5.0.3	axon-messaging-5.0.3	7	59	0.118644	[org.axonframework.messaging.core, org.axonframework.messaging.eventhandling.processing, org.axonframework.messaging.monitoring, org.axonframework.messaging.commandhandling, org.axonframework.messaging.queryhandling, org.axonframework.messaging.eventhandling, org.axonframework.messaging.monitori...	[core, processing, monitoring, commandhandling, queryhandling, eventhandling, configuration]
3	axon-metrics-micrometer-5.0.3	axon-common-5.0.3	2	15	0.133333	[org.axonframework.common.configuration, org.axonframework.common]	[configuration, common]
4	axon-test-5.0.3	axon-eventsourcing-5.0.3	1	7	0.142857	[org.axonframework.eventsourcing.eventstore]	[eventstore]
5	axon-server-connector-5.0.3	axon-eventsourcing-5.0.3	1	7	0.142857	[org.axonframework.eventsourcing.eventstore]	[eventstore]
6	axon-server-connector-5.0.3	axon-modelling-5.0.3	1	7	0.142857	[org.axonframework.modelling]	[modelling]
7	axon-metrics-micrometer-5.0.3	axon-spring-boot-autoconfigure-5.0.3	1	7	0.142857	[org.axonframework.extension.springboot.autoconfig]	[autoconfig]
8	axon-test-5.0.3	axon-messaging-5.0.3	9	59	0.152542	[org.axonframework.messaging.commandhandling, org.axonframework.messaging.core, org.axonframework.messaging.eventhandling, org.axonframework.messaging.core.conversion, org.axonframework.messaging.eventstreaming, org.axonframework.messaging.eventhandling.processing.streaming.token, org.axonframew...	[commandhandling, core, eventstreaming, conversion, token, annotation, monitoring]
9	axon-spring-boot-autoconfigure-5.0.3	axon-test-5.0.3	1	6	0.166667	[org.axonframework.test.server]	[server]
10	axon-modelling-5.0.3	axon-conversion-5.0.3	1	5	0.200000	[org.axonframework.conversion]	[conversion]
11	axon-conversion-5.0.3	axon-common-5.0.3	3	15	0.200000	[org.axonframework.common.infra, org.axonframework.common.annotation, org.axonframework.common]	[infra, annotation, common]
12	axon-eventsourcing-5.0.3	axon-conversion-5.0.3	1	5	0.200000	[org.axonframework.conversion]	[conversion]
13	axon-conversion-5.0.3	axon-conversion-5.0.3	1	5	0.200000	[org.axonframework.conversion]	[conversion]
14	axon-update-5.0.3	axon-common-5.0.3	3	15	0.200000	[org.axonframework.common.annotation, org.axonframework.common.configuration]	[annotation, common, configuration]
15	axon-spring-boot-autoconfigure-5.0.3	axon-server-connector-5.0.3	1	5	0.200000	[org.axonframework.axonserver.connector]	[connector]
16	axon-modelling-5.0.3	axon-messaging-5.0.3	13	59	0.220339	[org.axonframework.messaging.core.unitofwork, org.axonframework.messaging.eventhandling, org.axonframework.messaging.core, org.axonframework.messaging.commandhandling, org.axonframework.messaging.queryhandling.configuration, org.axonframework.messaging.commandhandling.configuration, org.axonfram...	[unitofwork, eventhandling, core, commandhandling, configuration, conversion, annotation, reflection]
17	axon-test-5.0.3	axon-common-5.0.3	4	15	0.266667	[org.axonframework.common, org.axonframework.common.configuration, org.axonframework.common.infra, org.axonframework.common.annotation]	[common, configuration, infra, annotation]
18	axon-server-connector-5.0.3	axon-messaging-5.0.3	16	59	0.271186	[org.axonframework.messaging.eventhandling.processing.subscribing, org.axonframework.messaging.eventstreaming, org.axonframework.messaging.eventhandling.processing.streaming, org.axonframework.messaging.core, org.axonframework.messaging.eventhandling.processing, org.axonframework.messaging.event...	[subscribing, eventstreaming, streaming, core, processing, token, conversion, unitofwork, segmentation, store, eventhandling, distributed, commandhandling, queryhandling]
19	axon-eventsourcing-5.0.3	axon-messaging-5.0.3	16	59	0.271186	[org.axonframework.messaging.eventhandling.annotation, org.axonframework.messaging.eventstreaming, org.axonframework.messaging.core.unitofwork, org.axonframework.messaging.eventhandling, org.axonframework.messaging.core, org.axonframework.messaging.eventhandling.processing.streaming.token, org.a...	[annotation, eventstreaming, unitofwork, eventhandling, core, token, conversion, commandhandling, configuration, interception, transaction]
20	axon-spring-boot-autoconfigure-5.0.3	axon-messaging-5.0.3	16	59	0.271186	[org.axonframework.messaging.core.timeout, org.axonframework.messaging.core.conversion, org.axonframework.messaging.core.unitofwork.transaction.jpaa, org.axonframework.messaging.core.annotation, org.axonframework.messaging.core, org.axonframework.messaging.eventhandling, org.axonframework.messaging...	[timeout, conversion, jpaa, annotation, core, eventhandling, correlation, interception, commandhandling, distributed, queryhandling, store, transaction]
21	axon-spring-boot-autoconfigure-5.0.3	axon-eventsourcing-5.0.3	2	7	0.285714	[org.axonframework.eventsourcing.eventstore.jpaa, org.axonframework.eventsourcing.eventstore]	[jpaa, eventstore]
22	axon-spring-boot-autoconfigure-5.0.3	axon-common-5.0.3	5	15	0.333333	[org.axonframework.common.annotation, org.axonframework.common.jpaa, org.axonframework.common.configuration, org.axonframework.common]	[annotation, jpaa, jdbc, configuration, common]
23	axon-modelling-5.0.3	axon-common-5.0.3	5	15	0.333333	[org.axonframework.common.infra, org.axonframework.common.configuration, org.axonframework.common.property, org.axonframework.common.annotation]	[infra, common, configuration, property, annotation]
24	axon-server-connector-5.0.3	axon-common-5.0.3	6	15	0.400000	[org.axonframework.common, org.axonframework.common.annotation, org.axonframework.common.configuration, org.axonframework.common.lifecycle, org.axonframework.common.util]	[common, annotation, configuration, infra, lifecycle, util]
25	axon-messaging-5.0.3	axon-conversion-5.0.3	2	5	0.400000	[org.axonframework.conversion, org.axonframework.conversion.jackson]	[conversion, jackson]
26	axon-common-5.0.3	axon-common-5.0.3	8	15	0.533333	[org.axonframework.common, org.axonframework.common.tx, org.axonframework.common.function, org.axonframework.common.annotation, org.axonframework.common.configuration, org.axonframework.common.lifecycle, org.axonframework.common.io]	[common, tx, function, annotation, configuration, infra, lifecycle, io]
27	axon-eventsourcing-5.0.3	axon-common-5.0.3	8	15	0.533333	[org.axonframework.common.infra, org.axonframework.common.configuration, org.axonframework.common, org.axonframework.common.jpaa, org.axonframework.common.tx, org.axonframework.common.jdbc, org.axonframework.common.io]	[infra, configuration, common, annotation, jpaa, tx, jdbc, io]
28	axon-spring-boot-autoconfigure-5.0.3	axon-update-5.0.3	3	5	0.600000	[org.axonframework.update.configuration, org.axonframework.update.detection, org.axonframework.update]	[configuration, detection, update]
29	axon-test-5.0.3	axon-test-5.0.3	4	6	0.666667	[org.axonframework.test, org.axonframework.test.util, org.axonframework.test.matchers, org.axonframework.test.fixture]	[test, util, matchers, fixture]

Table 7 - Types that are used by multiple artifacts

This table shows the top 30 types that only use a few (compared to all existing) types of another artifact. The whole table can be found in the CSV report `ClassesPerPackageUsageAcrossArtifacts`.

	artifactName	dependentArtifactName	packageName	dependentPackage.fqn	dependentTypes	dependentPackageTypes	typeUsagePercentage
0	axon-modelling-5.0.3	axon-messaging-5.0.3	org.axonframework.modelling.repository	org.axonframework.messaging.core	1	80	0.0%
1	axon-test-5.0.3	axon-messaging-5.0.3	org.axonframework.test.matchers	org.axonframework.messaging.core	1	80	0.0%
2	axonsourcing-5.0.3	axon-messaging-5.0.3	org.axonframework.eventsourcing.configuration	org.axonframework.messaging.core	1	80	0.0%
3	axonsourcing-5.0.3	axon-messaging-5.0.3	org.axonframework.eventsourcing.configuration	org.axonframework.messaging.core.annotation	1	51	0.0%
4	axonsourcing-5.0.3	axon-common-5.0.3	org.axonframework.eventsourcing.annotation	org.axonframework.common.configuration	1	46	0.0%
5	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.core.unitofwork	org.axonframework.common.configuration	1	46	0.0%
6	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.core	org.axonframework.common.configuration	1	46	0.0%
7	axon-modelling-5.0.3	axon-common-5.0.3	org.axonframework.modelling.entity.annotation	org.axonframework.common.configuration	1	46	0.0%
8	axonsourcing-5.0.3	axon-common-5.0.3	org.axonframework.eventsourcing.annotation.reflection	org.axonframework.common.configuration	1	46	0.0%
9	axon-metrics-micrometer-5.0.3	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.metrics.micrometer.springboot	org.axonframework.extension.springboot.autoconfig	1	41	0.0%
10	axon-tracing-opentelemetry-5.0.3	axon-messaging-5.0.3	org.axonframework.extension.tracing.opentelemetry	org.axonframework.messaging.core	2	80	0.0%
11	axon-conversion-5.0.3	axon-common-5.0.3	org.axonframework.conversion.avro	org.axonframework.common	1	34	0.0%
12	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.commandhandling.tracing	org.axonframework.common	1	34	0.0%
13	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.eventhandling.processing.streaming.token.store	org.axonframework.common	1	34	0.0%
14	axon-test-5.0.3	axon-common-5.0.3	org.axonframework.test.util	org.axonframework.common	1	34	0.0%
15	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.core.interception.annotation	org.axonframework.common	1	34	0.0%
16	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.core.unitofwork.annotation	org.axonframework.common	1	34	0.0%
17	axon-modelling-5.0.3	axon-common-5.0.3	org.axonframework.modelling.repository	org.axonframework.common	1	34	0.0%
18	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.commandhandling.interception	org.axonframework.common	1	34	0.0%
19	axon-tracing-opentelemetry-5.0.3	axon-common-5.0.3	org.axonframework.extension.tracing.opentelemetry	org.axonframework.common	1	34	0.0%
20	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.monitoring.configuration	org.axonframework.common	1	34	0.0%
21	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.eventstreaming	org.axonframework.common	1	34	0.0%
22	axon-test-5.0.3	axon-common-5.0.3	org.axonframework.test.server	org.axonframework.common	1	34	0.0%
23	axon-test-5.0.3	axon-common-5.0.3	org.axonframework.test.extension	org.axonframework.common	1	34	0.0%
24	axon-update-5.0.3	axon-common-5.0.3	org.axonframework.update	org.axonframework.common	1	34	0.0%
25	axonsourcing-5.0.3	axon-common-5.0.3	org.axonframework.eventsourcing.annotation	org.axonframework.common	1	34	0.0%
26	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.eventhandling.processing	org.axonframework.common	1	34	0.0%
27	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.eventhandling.tracing	org.axonframework.common	1	34	0.0%
28	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.monitoring	org.axonframework.common	1	34	0.0%
29	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.queryhandling.configuration	org.axonframework.common	1	34	0.0%

Table 8 - Duplicate package names across artifacts

This table shows the top 30 duplicate package names across artifacts. They are ordered by the number of duplicates descending.

This might lead to confusion, makes importing more error prone and might even lead to duplicate classes where only one of them will be loaded by the class loader. If a package is named the same way in two or more artifacts this even allows another artifact to access package protected classes, methods or members which might not be intended.

The whole table can be found in the CSV report `DuplicatePackageNamesAcrossArtifacts`.

packageName	duplicates	artifactNames
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Table 9 - Annotated elements

This table shows 30 most used Java Annotations including some examples where they are used.

	annotationName	languageElement	numberOfAnnotatedElements	
0	jakarta.annotation.NonNull	Parameter	3497	[org.axonframework.test.fixture.RecordingCommandBus.<init>(0), org.axonframework.test.fixture.org.axonframework.test.fixture.RecordingCommandBus.subscribe(0), org.axonframework.test.fixture.iorg.a
1	jakarta.annotation.NonNull	Method	865	[org.axonframework.test.extension.ProvidedAxonTestFixture.org.axonframework.test.fixture.ResourceParameterResolverFactory\$FailingParamorg.axonframework.common.util.ExecutorServiceFactory.createEx
2	jakarta.annotation.Nullable	Parameter	395	[org.axonframework.test.fixture.RecordingCommandBus.dispatch(1), org.axonframework.test.fixture.org.axonframework.test.fixture.AxonTestThenNothing.<init>(3), org.axonframework.test.fiorg.axonfram
3	jakarta.annotation.Nullable	Method	112	[org.axonframework.test.fixture.RecordingCommandBus.resultOf, org.axonframework.test.extension.AxonFraorg.axonframework.common.ReflectionUtils.declaringClass, org.axonframeworkorg.axonframework.co
4	org.axonframework.common.annotation.Internal	Class	83	[org.axonframework.test.fixture.RecordingComponentsRegistry, org.axonframeworkorg.axonframework.test.fixture.RecordingEventSink, org.axonframework.test.fixture.Messaorg.axonframework.test.util.Rec
5	jakarta.annotation.NonNull	Field	72	[org.axonfrCustomization.fieldFilters, org.axonframework.common.configuration.AbstractComrg.axonframework.common.configuration.Component\$Identifier.type, org.axonframework.eventsourcing.eve
6	java.lang.FunctionalInterface	Interface	59	[org.axonframework.test.extension.AxonTestFixtureProvider, org.axonframework.test.matchers.FieldFilter, org.a>SqlResultConverter, org.axonframework.common.jdbc.ConnectionProvider, org.axonframeSqlFunctionio
7	org.springframework.context.annotation.Bean	Method	45	[org.axonframework.extension.metrics.micrometer.springboot.MicrometerMetricsAutoConfiguration.dorg.axonframework.extension.metrics.micrometer.springboot.MicrometerMetorg.axonframework.exten!
8	java.lang.annotation.Target	Annotation	42	[org.axonframework.test.extension.ProvidedAxonTestFixture, org.axonframework.common.annotation.Internal,org.axonframework.eventsourcing.annotation.EventSourcedEntity, org.axonframeworkorg.axor
9	java.lang.annotation.Retention	Annotation	42	[org.axonframework.test.extension.ProvidedAxonTestFixture, org.axonframework.common.annotation.Internal,org.axonframework.eventsourcing.annotation.EventSourcedEntity, org.axonframeworkorg.axor
10	org.springframework.boot.autoconfigure.condition.ConditionalOnMissingBean	Method	23	[org.axonframework.extension.metrics.micrometer.springboot.MicrometerMetorg.axonframework.extension.metrics.micrometer.springboot.MicrometerMetricsAutoConfigurorg.axonframework.extension.sq
11	org.springframework.boot.autoconfigure.AutoConfiguration	Class	21	[org.axonframework.extension.metrics.micrometer.springboot.Iorg.axonframework.extension.springboot.autoconforg.axonframework.extension.springboot.autoconfig.AxonTimeoutAutoConfiguration, org.a
12	org.axonframework.common.annotation.Internal	Interface	18	[org.axonframework.common.jdbc.ConnectionProvider, org.axonframeworkorg.axonframework.common.jpa.EntityManagerProvider, org.axonframeworkorg.axonframework.eventsourcing.eventstore.EventCoordinator,
13	org.axonframework.common.annotation.Internal	Constructor	17	[org.axonframework.eventsourcing.eventstore.InterceptingEventStore.<init>, org.axonframework.converterorg.axonframework.conversion.jackson.JacksonConverter.<init>, org.axonframework.ciorg.axor
14	java.lang.annotation.Documented	Annotation	15	[org.axonframework.common.annotation.Internal, org.axonframework.eventsourciorg.axonframework.messaging.commandhaorg.axonframework.messaging.eventhandling.annotation.EventHandler, org.axonframe
15	com.fasterxml.jackson.annotation.JsonProperty	Parameter	12	[org.axonframework.messaging.eventhandling.processing.streaming.tokenorg.axonframework.messaging.eventhandling.processing.streaming.tokenorg.axonframework.messaging.eventhandling.processing.strei
16	org.axonframework.common.Priority	Class	12	[org.axonframework.test.Fixturorg.axonframework.messaging.eventhandling.annotation.Sequenorg.axonframework.messaging.eventhandling.annotation.TimestampParameterResolverFactory, org.axonframework
17	org.springframework.boot.autoconfigure.condition.ConditionalOnProperty	Class	12	[org.axonframework.extension.springboot.autoconfig.AvroSchemaStoreAutoConfigurationAvroConfiguredorg.axonframework.extension.springboot.autoconfig.Avro
18	org.axonframework.common.annotation.Internal	Method	11	[org.axonframework.messaging.eventhandling.conversionorg.axonframework.messaging.eventhandling.processing.streaming.tokenorg.axonframework.messaging.eventhandling.processing.streaming.token
19	org.springframework.boot.context.properties.EnableConfigurationProperties	Class	11	[org.axonframework.extension.metrics.micrometer.springboot.Iorg.axonframework.extension.springboot.autoconfigorg.axonframework.extension.springboot.autoconfig.Jackson2MapperAutoConfigur, org.ax
20	org.springframework.boot.context.properties.ConfigurationProperties	Class	10	[org.axonframework.extension.metrics.micrometer.springboot.MetricsProperties, org.axonframework.axonserveiorg.axonframework.extension.springboot.DistributedCommandBusProperties, org.axonframework.exten!
21	org.springframework.boot.autoconfigure.condition.ConditionalOnClass	Class	9	[org.axonframework.extension.metrics.micrometer.springboot.Iorg.axonframework.extension.springboot.autoconfiorg.axonframework.extension.springboot.autoconfig.AvroSchemaStoreAutoConfiguration, or
22	jakarta.persistence.Basic	Field	8	[org.axonframework.eventsourcing.evenorg.axonframework.eventsourcing.eventstorg.axonframework.eventsourcing.eventstore, org.axonframework.eventsourcing.evi
23	org.springframework.boot.autoconfigure.AutoConfigureBefore	Class	8	[org.axonframework.extension.metrics.micrometer.springboot.Iorg.axonframework.extension.springboot.autoconfig.Correlorg.axonframework.extension.springboot.autoconfig.Jackson2MapperAutoConfigur
24	org.springframework.boot.autoconfigure.condition.ConditionalOnProperty	Method	8	[org.axonframework.extension.metrics.micrometer.springboot.MicrometerMetricsAutoConfiguration.dorg.axonframework.extension.metrics.micrometer.springboot.MicrometerMetorg.axonframework.exten!
25	org.springframework.boot.context.properties.NestedConfigurationProperty	Field	8	[org.axonframework.extMessageHandlerTimeoutProperties.events, org.axonframework.extension.MessaMessageHandlerTimeoutProperties.commands, org.axonframework.extension.springboot.TimeoutProperties\$Messa
26	java.lang.SafeVarargs	Constructor	7	[org.axonframework.test.matchers.SequenceMatcher.<init>, org.axonframeworkorg.axonframework.test.matchers.ExactSequenceMatcher.<init>, org.axonframework.test.nrorg.axonframework.test.matchers
27	java.lang.Deprecated	Method	7	[org.axonframework.test.fixture.AxonTestThenCommand.resultMessagePayloadSatisfies, org.axonframework.core.GenCommand.resultMessagePayloadSatisfies, org.axonframework.messaging.core.Genorg.axonframework.mes
28	org.springframework.context.annotation.Conditional	Method	7	[org.axonframework.extension.springboot.autoconfig.Jackson2MapperAutoConfigorg.axonframework.extension.springboot.autoconfig.AvroSchemaStoreAutoCoiorg.axonframework.extension.springboot.autoconfi
29	com.fasterxml.jackson.annotation.JsonCreator	Constructor	6	[org.axonframework.messaging.eventhandling.processing.streaming.tokenorg.axonframework.messaging.eventhandling.processing.streaming.streaiorg.axonframework.messaging.eventhandling.processing.streaming.token

Table 10 - Distance distribution between dependent files

This table shows the file directory distance distribution between dependent files. Intuitively, the distance is given by the fewest number of change directory commands needed to navigate between a file and a dependency it uses. Those are aggregate to see how many dependent files are in the same directory, how many are just one change directory command apart, and so on.

dependency.fileDistanceAsFewestChangeDirectoryCommands	numberOfDependencies	numberOfDependencyUsers	numberOfDependencyProviders	
0	0	2125	881	921
1	1	100	86	41
2	2	2138	631	420
3	4	2104	679	322